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Quantifying age-related differences in visual-discrimination capacity: Drivers with and without visual impairment

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ABSTRACT

The aim of this study is to examine the effects of aging as well as visual impairment on retinal-image quality and visual performance in drivers. We use a new visual test called Halo v1.0 software for quantifying the discrimination capacity, an important visual function for evaluating the visual disturbances perceived by the observer. The study included 55 subjects with normal vision and 15 older subjects with cataracts. All subjects were examined for visual acuity, contrast sensitivity, visual-discrimination capacity and optical quality. Subjects also completed a subjective Driving Habits Questionnaire (DHQ). Older drivers with and without visual impairment showed significantly ($p < 0.05$) worse visual performance and deteriorated retinal-image quality, even when their binocular visual acuity was $\geq$20/25. In conclusion, some visual functions are considerably diminished in older drivers, even when visual acuity is sufficient to get or renew a driver's license. Halo software enables easy quantification of night-vision disturbances such as halos, which could impede the detection of pedestrians, cyclists, or traffic signals, thereby making this system advisable in clinical practice, e.g. in the requirements for a driver's license, particularly for older drivers.

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1. Introduction

The quality of retinal images is affected by such defects as aberrations and scattering, caused by the ocular characteristics themselves or by different pathologies. With age, changes arise in the different eye structures, and the loss of transparency causes a physiological increase in ocular scattering, mainly at the level of the lens, although the cornea and the vitreous humour also contribute to this condition to a certain degree (Pierscionek, 1996). When the loss of lens transparency worsens, it can trigger the formation of cataracts. Cataracts are the major cause of blindness and visual impairment in developing countries (Rao et al., 2011). A study in the United States showed that cataracts constitute the leading cause of bilateral vision worse than 20/40 among white, black and Hispanic persons (Congdon et al., 2004). Cataracts are frequently present in drivers over 65 years of age (Nischler et al., 2010). This visual pathology diminishes visual acuity and contrast sensitivity (Babizhayev, 2003; Mäntyjärvi and Tuppurainen, 1999). In driving, decreased contrast sensitivity lowers the discrimination capacity in such a way that pedestrians, cyclists and traffic signs could go undetected by the driver, particularly at dusk or under unfavourable meteorological conditions such as fog, rain, or snowfall. This can compromise traffic safety in older drivers affected with cataracts (Babizhayev, 2003; Owsley et al., 1999).

It has been suggested that the older drivers with and without visual impairment were significantly less safe than the younger and middle-aged drivers (Babizhayev, 2003; Owsley et al., 1999; Ryan et al., 1998; Wood and Mallon, 2001). As the number of elderly is increasing, the number of older drivers will also rise in the coming years.

Often, persons with some visual pathology or deterioration manage to get a driver's license if their binocular vision is 20/40 or better. The test most frequently used in the clinic to predict visual function is the measurement of visual acuity; however, in visual functions, there are other factors that can affect traffic safety, such as contrast sensitivity, dark adaptation, colour vision, glare sensitivity, and visual field (Mäntyjärvi and Tuppurainen, 1999; Mäntyjärvi et al., 1999; Rogé and Pébayle, 2009; Van Rijn et al., 2011). It has been recommended that a contrast-sensitivity test be added for driver's license testing in older people (Decina and Staplin, 1993). However, psychophysical tests are time consuming and constitute a difficult task for older people. On the other hand, the use of questionnaires has serious limitations because the subjects may fail to reveal the magnitude of their problem. For these reasons, we consider it important to include standardized

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tests in clinical practice that quickly and reliably characterize the critical aspects of vision needed in traffic in order to determine whether a person fulfils the requisites to get or renew a driver's license, particularly for older drivers.

The aim of this study was to examine the way in which the normal aging process, as well as eye disease such as cataracts, affect optical quality and visual performance in younger and older drivers (with and without visual impairment), and to find out whether other visual functions, apart from visual acuity, should be taken into account in the issuing of a driver's license. We use a new visual test for quantifying visual-discrimination capacity (Castro et al., 2011), this being an important visual function for quantifying the visual disturbances, such as halos, that affect visual performance (Gutiérrez et al., 2003; Jiménez et al., 2006; Villa et al., 2007) and could hamper tasks such as night driving, especially in older drivers.

2. Methods

2.1. Participants

A total of 70 participants were recruited for this study. Of these, 55 subjects without any ocular disease and having best-corrected visual acuity equal to or better than 20/20 in both eyes were divided into three age categories: groups of 20 younger (19–29 years old, mean age 24.7 $\pm$ 2.6 years, standard deviation included); 20 middle-aged (30–59 years old, mean age 44.1 $\pm$ 7.7 years); and 15 older subjects (>60 years old, mean age 68.5 $\pm$ 4.4 years). Also included were 15 subjects older than 60 years of age and with a clinically relevant cataract in at least one eye (>60 years old, mean age 71.4 $\pm$ 5.0 years) with binocular visual acuity equal to or better than 20/25. The cataract was classified as nuclear in 13 subjects (26 eyes) and posterior subcapsular in 2 subjects (4 eyes). Each cataract subject was classified according to the degree of nuclear opacification (NO), nuclear colour (NC), and posterior subcapsular opacification (P), by using the LOCSIII chart (Chylack et al., 1993). The 30 cataract eyes were classified as follows: 16 eyes as grade 1 (NO1, NC1), 10 eyes as grade 2 (NO2, NC2), and 4 eyes as P1. There were no significant differences in age between the older subjects without any ocular diseases and those affected with cataracts.

All participants were licensed drivers and reported that they drove regularly.

The cataract group was diagnosed by the same ophthalmologist at the Adeslas S.A. Clinic of Granada (Spain). All participants in the study gave their informed consent in accordance with the Helsinki Declaration.

2.2. Study design and procedures

Snellen visual acuity was measured monocularly and binocularly for all participants with their best correction at a working distance of 6 m. The pupil diameter was measured with a Colvard pupillometer (OASIS, Glendora, CA, USA) to ensure that all patients during visual performance measurements (the contrast-sensitivity function and the discrimination capacity) reached a pupil diameter greater than 4 mm. The patients were examined binocularly for the visual performance (the contrast-sensitivity function and the discrimination capacity), because the binocular visual acuity required for a driver's license is $\geq$20/40.

To determine the optical quality, we took objective data from a visual-quality device, Optical Quality Analysis System (Visiometrics SL, Tarrasa, Spain), based on the double-pass technique (Díaz-Doutón et al., 2006; Güell et al., 2004). The following optical-quality parameters were analysed: the cut-off frequency of the modulation transfer function (MTF), the Strehl ratio, and the objective scattering index (OSI).

The participants also completed a subjective Driving Habits Questionnaire (DHQ), designed to compile information about their driving during the past year (Owsley et al., 1999). This questionnaire is frequently used in these types of studies (Babizhayev et al., 2009; Ball et al., 1998; McGwin et al., 2000).

2.3. Contrast-sensitivity function

The CSF test was administered with a B-VAT II device (Mentor O&O, Inc.) using the following spatial frequencies: 1.5, 2.4, 3.7, 6.0, 9.2, 12.0, 15.0, 20.0, and 24.0 cycles per degree (cpd). The test was administered binocularly at 6 m corresponding to a visual angle of 1.8°. The mean luminance for all the frequencies was within the mesopic range (in all cases 4 cd/m^2), to simulate the conditions during night driving when visual performance is diminished. Sinusoidal gratings may be oriented vertically or 14° clockwise or anticlockwise from vertical and they are presented randomly, to prevent memorization.

In each subject, the CSF was determined using the method of the limits (Anera et al., 2003; Jiménez et al., 2003, 2006) as follows: the subject, was asked to indicate the orientation of the grating from among 3 possibilities, beginning with the lowest spatial frequency and the highest contrast value available. This contrast value was reduced until the subject was unable to answer correctly. The procedure was repeated 3 times. The threshold contrast was the mean of the 3 contrasts perceived by the subject. Once the contrast threshold had been determined, the procedure was repeated using a larger spatial-frequency grating.

CSF data were taken for all participants with their best correction for the distant visual acuity. Before the data were recorded, each observer underwent several training sessions to ensure the correct understanding of the test.

2.4. Discrimination-capacity measurements

For a complete evaluation of the visual performance, it is important to quantify the discrimination capacity, which is deteriorated in the presence of visual disturbances perceived by the observer, especially in subject diagnosed with an ocular pathology such as cataract. A deterioration in the discrimination capacity can affect certain tasks such as night-time driving, hampering visibility of pedestrians or the recognition of traffic signals. To evaluate the discrimination capacity, we sought to quantify the visual disturbances perceived by the subject under low-illumination conditions. For this, we used a new visual test called Halo v1.0 software (*Laboratory of Vision Sciences and Applications*, University of Granada, Spain). The subject's task consisted of discriminating luminous peripheral stimuli around a central high-luminance stimulus over a dark background. More information on this visual test can be found elsewhere (Castro et al., 2011). Halo software is freeware and, free of charge, can be downloaded from the laboratory webpage: http://www.ugr.es/~labvisgr/.

2.4.1. Measurement of visual-disturbance index

After the test was finished, the visual-disturbance index was calculated as a ratio between non-detected stimuli and all the peripheral stimuli presented to the observer according to Eq. (1). This ratio takes into account the distance of each undetected peripheral stimulus to the main stimulus centre, adjusted for the times that the corresponding peripheral stimulus was not detected by the subject. The distance dependence is also present in the denominator, where all the peripheral stimuli are considered. The disturbance index is calculated by Eq. (1):

$$\rho_q = \left(\Sigma_{i=1}^{N} p_i r_i^2\right) \Big/ \left(p \Sigma_{i=1}^{N} r_i^2\right) \quad (1)$$

where r_i is the distance (in pixels) from the centre of the central stimulus to the centre of the i-peripheral stimulus, considering a concrete semiaxis; N is the total number of peripheral stimuli; p is the total weight (number of times that each stimulus i is shown); and after being selected is the same for all the peripheral stimuli; and p_i is the number of times over the total weight ($pi \leq p$) that the i-peripheral stimulus is not detected by the subject.

The disturbance index takes values from 0 to 1. The greater value of this parameter indicates that there is a greater contribution of visual disturbances, such as glare, a veil of stray light over the retinal image, or visual halos around luminous stimuli, and therefore poorer discrimination capacity.

In addition to the disturbance index, the Halo software generates a graph of results, showing areas where the peripheral stimuli were not detected by the observer and areas where the peripheral stimuli were detected totally or partially. In the graph, the central stimulus is shown as well as the number of times that each peripheral stimulus is detected by the observer (X for being undetected, or 1, 2, or 3 if detected one, two or three times, respectively), with respect the total weight ($p = 3$). These values (X, 1, 2, or 3) are placed in the corresponding position where each peripheral stimulus was shown. This graph describes the shape of the visual disturbances perceived by the observer (halos, starbursts or glare), showing information about areas around a high-luminance stimulus where the observer presents difficulties on detecting different luminous stimulus, and therefore giving complete information on the discrimination capacity.

2.4.2. Experiment arrangement

The discrimination capacity was assessed using an LCD-monitor with the resolution set at 1024 × 768 pixels. The distance from the observer to the monitor was 2.5 m and the test was performed binocularly, the normal state in driving. The size of the stimuli was 30 pixels for the radius of the central stimulus and 1 pixel for the radius of the peripheral stimuli, subtending 0.46 and 0.02°, respectively (configuration 1). The luminance of the stimuli was measured with a spectroradiometer SpectraScan PR-650 (Photo Research, Incorporated, Chatsworth, CA, USA), being of 176.1 cd/m^2 for the main stimulus and 61.1 cd/m^2 for the peripheral one, with the luminance for the background monitor of 0.71 cd/m^2. The monitor showed 72 peripheral stimuli around the central one, distributed along 18 semiaxes (four stimuli per semiaxis) in order to evaluate a higher area around the central stimulus. The maximum radius of each semiaxis was 60 pixels (the most distant stimulus being 60 pixels from the centre of the main stimulus). Configuration 1 was used for both groups (control and cataracts). However, with the configuration used, 6 cataract subjects with a greater visual deterioration found it difficult to detect peripheral stimuli, failing to detect any of the stimuli presented. Therefore, for these 6 subjects, the results with this configuration showed the maximum value for the disturbance index (disturbance index = 1), thereby losing information in the discrimination capacity. To achieve an accurate disturbance index, we increased the size of the peripheral stimuli from 1 to 2 pixels, subtending 0.04° from the observer's position (configuration 2). In this case, the maximum radius was 70 pixels, to compensate for the increment in the peripheral stimuli size and to extend the region of the stimulus detection. The rest of the parameters were equal for all subjects. This configuration agrees with spatial parameters used in other works (Castro et al. 2011; Gutiérrez et al., 2003; Jiménez et al., 2006).

We used a weight of 3 ($p = 3$), taking p_i values of 0 (peripheral stimulus detected) or 1, 2, or 3 (stimulus not detected one, two, or three times, respectively). Also, we used a weight of 3 to assess a higher level of accuracy. Therefore, the monitor showed a total of 216 peripheral stimuli around the central one.

The exposure time of each peripheral stimulus was 1 s, a value closer to the real night-driving situations where the subject must discriminate or distinguish lights or pedestrians. After the exposure of a peripheral stimulus, a time lapse between the stimuli (refresh) of 0.8–2 s was used. During this time, only the central spot was presented.

To simulate night-driving situations such as the headlights of an oncoming car in which the subjects could indicate night-vision disturbances (glare or halos), the measurements were performed with the examining room dark and the minimum background luminance of the monitor to provide high contrast with the stimulus. The illuminance at the observer position with the halo test running was of 0.03 lux, measured with an illuminance meter T-10 (Konica Minolta Sensing, Inc., NJ, USA).

2.4.3. Test protocols

Before the formal session began, an initial trial was executed to check the subject's correct understanding of the test. The formal testing began after the observer's position was fixed in front of the monitor with a chin and forehead rest. A session was conducted as follows: after a 3-min adaptation period to darkness of the monitor background, there was 1-min adaptation to the main stimulus, and then the subject was randomly presented with peripheral stimuli around to the central stimulus, to avoid learning effects. The patient, on detecting peripheral spots, pressed a button on the mouse, storing this information for subsequent treatment and calculation of the disturbance index.

2.5. Optical-quality measurements

To determine the optical quality, we took objective data from an optical device based on the double-pass technique (Díaz-Doutón et al., 2006; Güell et al., 2004). The Optical Quality Analysis System [(OQAS), Visiometrics SL, Tarrasa, Spain], provides data on diffraction, ocular aberrations, and scattering that diminish the image clarity, reducing visual quality of the subject. This objective optical device is useful in patients having an ocular pathology such as cataract or in older patients for whom the influence of scattering could be great due to natural changes (mainly in the lens) caused by aging. Because of this, we found a worse retinal-image quality and a higher number of difficulties often encountered during driving (glare, signal recognition, and visibility of the pedestrians).

2.5.1. The modulation transfer function cut-off frequency (MTF cut-off), Strehl ratio and objective scattering index (OSI)

The MTF represents the loss of contrast produced by the eye's optics on a sinusoidal grating as a function of its spatial frequency (Díaz-Doutón et al., 2006). We analysed the MTF cut-off for a complete evaluation of the ocular optical quality. Theoretically, this parameter represents the spatial frequency, in cycles per degree (cpd), corresponding to an MTF value of 0. However, due to the noise introduced by the CCD camera in the OQAS device, an MTF value of 0.01 is considered in order to determine the MTF cut-off. A higher value of the MTF cut-off indicates a better optical quality of the eye. On the other hand, we took the Strehl ratio, defined as the ratio between the 2D-MTF area of the eye and the diffraction-limited 2D-MTF area. The Strehl ratio ranges from 0 to 1. A lower value of this parameter indicates a greater contribution of aberrations and ocular scattering and therefore poorer optical quality. Finally, we also took the OSI, the only parameter that permits the objective quantification of the intraocular scattering. For younger eyes, the OSI value is lower than 0.5; for eyes with an

early cataract the OSI value ranges from 1.4 to 4 and, while for eyes with a cataract the value is higher to 4. More information on this device can be found elsewhere (e.g. Díaz-Doutón et al., 2006; Güell et al., 2004).

2.5.2. Test protocols

The data from the double-pass device were referred to a pupil size of 4 mm for all subjects and these data were taken with no dilation in order to maintain natural conditions. With the OQAS, we took at least 3 measurements in each eye for each parameter (the MTF cut-off, Strehl ratio and OSI) and averaged them. The measurements were made with the examining room dark.

2.6. Assessing driving performance

The Driving Habits Questionnaire (DHQ) was designed to gather information about driving during the past year (Owsley et al., 1999). Several items rate the degree of visual difficulty experienced in specific driving situations. Ratings were made on a 5-point scale (5 = no difficulty, 4 = a little difficulty, 3 = moderate difficulty, 2 = extreme difficulty, 1 = so difficult that I no longer drive in that situation).

2.7. Data analysis

Statistical analysis was performed by Student's *t*-test with $p = 0.05$ taken as the upper limit of significance. *t*-test was used to examine differences between two independents groups (older drivers with and without cataracts). For CSF, quantitative variables were compared using a bifactorial analysis of variance, with type of older driver (with and without cataracts) as the main factors. In addition to the bifactorial analysis, a *post hoc* comparison (the Bonferroni test, with a minimum level of significance of 0.05) was made for each spatial frequency considered in order to compare means for the drivers individually. We also used an analysis of variance to test for differences between groups (younger, middle-aged, and older) and parameters measured (disturbance index, Strehl ratio, OSI and MTF cut-off) as the dependent variables.

3. Results

3.1. Control group

According to the retinal-image quality, we found a lower value of the Strehl ratio and a higher OSI as age increased (Table 1), indicating a worse retinal-image quality due to a combination of greater intraocular scattering, aberrations, and retinal reflection. The results show significant differences in the Strehl ratio ($p = 0.009$) and in the OSI ($p = 0.013$) among the group of young subjects (19–29 years of age) and the group of more than 60 years of age. This deterioration was reflected in the MTF mean cut-off found for each group, which was significantly lower ($p < 0.05$) for older drivers. For the discrimination capacity, we found similar results: a higher disturbance index with age (Table 1), this being statistically significant between the younger group (19–29 years old) and the group older than 60 years of age ($p = 0.015$). The higher disturbance index indicates a deteriorated discrimination capacity resulting from the many changes that occur within the lens with the age, such as increased intraocular scattering. As an example of this deterioration, Fig. 1 shows the graphs pertaining to the Halo software for three different drivers: a younger, a middle-aged, and an older subject. We can see that the older driver presents a greater number of undetected peripheral stimuli and a higher disturbance index ($\rho_q = 0.10$, $\rho_q = 0.26$, and $\rho_q = 0.42$ respectively). In addition, the figure reflects that, as the age of the driver increases, the number of undetected peripheral stimuli around the central stimulus increase, that is, the area corresponding to these undetected stimuli augments.

Fig. 2 presents the results for binocular contrast-sensitivity function (as a function of the spatial frequencies considered) for the average of 20 younger, 20 middle-aged, and 15 older subjects without any ocular pathology (control group). This figure clearly reflects the deterioration in visual performance (lower contrast-sensitivity function) for older subjects. An ANOVA shows that this deterioration in older subjects was significant ($p < 0.05$) with respect younger and middle-aged groups for at least 5 spatial frequencies: 2.4; 3.7; 6.0; 12.0, and 15.0 cpd (Table 2).

Table 3 indicates the results for each driving situation with respect to the age of the subject. We found less difficulty for the older group although there were no significant differences except with regard to parallel parking between the younger and the older group ($p = 0.031$). These differences could have arisen because the subjective questionnaire has serious limitations in the sense that subjects may fail to reveal the full magnitude of the problem or may not be aware of the vision loss until reaching an advanced state of visual deterioration, considering their way of driving to be normal and safe. For this reason, could be recommendable to examine others visual functions by implementing in clinical practice, quick, easy, and reliable tests that provide a complete evaluation of the visual performance. This would make the subject aware for any limitations and thereby increase safety in everyday tasks such as driving.

3.2. Cataract group

For older drivers (with and without cataracts), we examined the retinal-image quality. For subjects with cataracts, the Strehl ratio ranged from 0.04 to 0.25 with a mean value of 0.12 ± 0.05 (standard deviation). In the case of older drivers without cataracts the mean was 0.17 ± 0.05, indicating a significantly lower value ($p < 0.001$) in the Strehl ratio for drivers affected with cataracts (Table 4). Regarding the OSI, drivers with cataracts registered a mean value of 2.91 ± 1.55 (ranging from 0.5 to 7.0) and 1.15 ± 0.51 for older drivers without cataracts, the differences being significantly higher for the cataract group ($p < 0.001$). The data reflect a clearly higher influence of aberrations and intraocular scattering for drivers affected with cataracts. We found a mean MTF cut-off of 18.56 ± 10.86 cpd for drivers having cataracts, this being significantly lower ($p < 0.001$) than the MTF cut-off found for healthy older drivers (29.67 ± 10.73 cpd) (Table 4).

In addition, we examined the discrimination capacity for older drivers (with and without cataracts) excluding 6 subjects who had cataracts, using a different spatial configuration for these subjects. For the initial parameters (configuration 1), the discrimination

Table 1
Data for the optical quality and discrimination capacity. Standard deviation included.

Age	Strehl ratio mean ± SD (range)	OSI mean ± SD (range)	MTF cut-off (cpd) mean ± SD (range)	Disturbance index mean ± SD (range)
19–29 years	0.24 ± 0.07 (0.13–0.44)	0.57 ± 0.32 (0.20–1.20)	40.29 ± 9.38 (18.21–56.91)	0.15 ± 0.08 (0.04–0.39)
30–59 years	0.20 ± 0.06 (0.09–0.34)	0.85 ± 0.65 (0.20–3.00)	34.11 ± 12.04 (9.54–54.87)	0.18 ± 0.08 (0.03–0.36)
>60 years	0.17 ± 0.04 (0.10–0.26)	1.11 ± 0.50 (0.30–2.30)	29.67 ± 10.73 (14.35–51.68)	0.23 ± 0.08 (0.13–0.42)

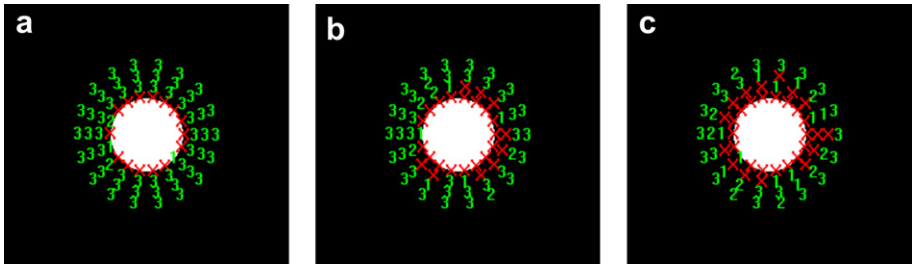


Fig. 1. Graphs made with the Halo v1.0 software for three different drivers: (a) a younger driver ($\rho_q = 0.10$), (b) a middle-aged driver ($\rho_q = 0.26$), (c) an older driver without cataracts ($\rho_q = 0.42$).

capacity in these 6 subjects with cataracts was clearly lower than in the rest of the subjects, given that the maximum value of 1 for the disturbance index was reached; for this reason, we increased the size of the peripheral stimuli. According to a statistical analysis, there were significant differences ($p = 0.010$) for older drivers (with and without cataracts) in the disturbance index (Table 4). The data from the software indicated an impairment of the discrimination capacity in subjects affected with cataracts. As an example, Fig. 3 shows the graphs pertaining to the Halo software for an older driver without cataracts and an older driver with cataracts. We found that a higher number of stimuli were undetected by the driver affected with the pathology and a higher disturbance index ($\rho_q = 0.42$, and $\rho_q = 0.81$ respectively). The figure reflects that the driver affected by the pathology presented difficulties in detecting not only the stimuli that were close to the central stimulus but also those that were farther away, thus expanding the area corresponding to the undetected stimuli. This was due to a higher level of ocular scattering that contributes to the sensation of glare as well as to the perception of larger halos around central lights, severely diminishing the capacity to detect peripheral lights surrounding the central source.

Fig. 4 presents the results for binocular contrast-sensitivity function (as a function of the spatial frequencies considered) for the older subjects (with and without cataracts). This figure clearly reflects the deterioration in visual performance (lower contrast-sensitivity function) for subjects affected with the pathology. The results in the binocular contrast-sensitivity function were significantly worse for subjects with cataracts than for cataract-free subjects in four spatial frequencies: 6.0, 9.2, 12.0, and 24.0 cpd (Table 5).

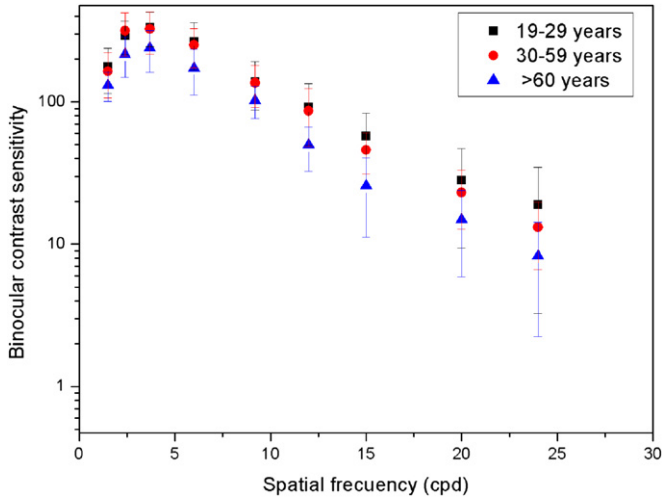


Fig. 2. Average binocular CSF corresponding to the 20 younger, 20 middle-aged and 15 older subjects without any ocular pathology. Data include standard error.

With respect to driving difficulty, older subjects with and without cataracts were scored as 0 "no difficulty" *vs.* 1 "difficulty". Table 6 illustrates the results for each driving situation. In the cataract group, the greater driving difficulty proved significant ($p < 0.05$) with respect to the no-cataract group in parallel parking, driving in heavy traffic, driving in rush hour, and driving at night.

4. Discussion

With age, changes occur in the different ocular structures, one of the most frequent being transparency loss in the lens, which can give rise to the formation of cataracts. These changes, whether a consequence of aging itself or due to an eye disease, increase ocular scattering. This greater scattering is reflected in a worsening of retinal-image quality and contributes to the sensation of glare as well as to the perception of halos around central lights, diminishing the contrast sensitivity and the ability to detect peripheral stimuli (reducing discrimination capacity), and therefore diminishing the overall visual performance of the subject, especially under low-illumination conditions. This deterioration in visual performance could cause trouble in seeing pedestrians and

Table 2
Data for the average of 20 younger, 20 middle-aged and 15 older drivers without any ocular pathology corresponding to binocular CSF.

Spatial frequencies (cpd)	Range of age (years)	CSF mean ± SD	*p*-value
1.5	19–29	176.89 ± 62.04	>60
	30–59	163.92 ± 57.04	19–29 (0.049)
	>60	131.31 ± 30.31	30–59 (0.235)
2.4	19–29	292.56 ± 77.79	>60
	30–59	316.69 ± 107.00	19–29 (0.049)
	>60	217.39 ± 68.09	30–59 (0.005)
3.7	19–29	335.13 ± 90.98	>60
	30–59	324.19 ± 107.33	19–29 (0.015)
	>60	239.57 ± 77.94	30–59 (0.034)
6.0	19–29	265.20 ± 96.69	>60
	30–59	251.31 ± 77.79	19–29 (0.006)
	>60	173.48 ± 61.05	30–59 (0.021)
9.2	19–29	139.65 ± 52.15	>60
	30–59	136.04 ± 44.27	19–29 (0.052)
	>60	102.83 ± 26.16	30–59 (0.088)
12.0	19–29	91.83 ± 42.52	>60
	30–59	86.68 ± 38.00	19–29 (0.004)
	>60	49.89 ± 17.15	30–59 (0.011)
15.0	19–29	57.37 ± 26.23	>60
	30–59	46.00 ± 14.92	19–29 (0.000)
	>60	25.82 ± 14.59	30–59 (0.012)
20.0	19–29	28.22 ± 18.82	>60
	30–59	23.02 ± 10.24	19–29 (0.020)
	>60	14.91 ± 8.99	30–59 (0.264)
24.0	19–29	18.96 ± 15.69	>60
	30–59	13.20 ± 6.56	19–29 (0.016)
	>60	8.28 ± 6.03	30–59 (0.542)

Table 3
Driving-difficulty questionnaire with respect to the age of the subject. Data presented as percentages.

Driving habits questionnaire	19–29 years (% of group)	30–59 years (% of group)	>60 years (% of group)
1. Driving in the rain			
Difficulty	53	55	20
No difficulty	47	45	80
2. Driving alone			
Difficulty	5	0	0
No difficulty	95	100	100
3. Parallel parking			
Difficulty	37	25	0
No difficulty	63	75	100
4. Left turns in traffic			
Difficulty	16	15	13
No difficulty	84	85	87
5. Driving on freeways			
Difficulty	0	0	0
No difficulty	100	100	100
6. Driving in heavy traffic			
Difficulty	26	35	20
No difficulty	74	65	80
7. Driving in rush hour			
Difficulty	26	40	20
No difficulty	74	60	80
8. Driving at night			
Difficulty	50	55	53
No difficulty	50	45	47

road signs, this representing a risk factor in traffic. According to some studies, older drivers with and without visual impairment are significantly less safe than the younger and middle-aged drivers (Babizhayev, 2003; Owsley et al., 1999; Ryan et al., 1998; Wood and Mallon, 2001).

According to our results, older drivers without cataracts showed a more deteriorated retinal-image quality and a lower discrimination capacity under low-illumination conditions (Table 1 and Fig. 1), which corresponds to a decline in the binocular CSF (Table 2 and Fig. 2). These results imply that older drivers could have greater difficulty in detecting peripheral stimuli such as pedestrians, cyclists, and road signs and therefore could have less ability to undertake certain activities such as driving, particularly at night. However, the subjective data compiled from the DHQ interview, reflect a lower difficulty for the older group although there were no significant differences except with regard to parallel parking between the younger and the older group. The reason could be that objective evidence of driving ability (performance or event history) does not appear to affect a driver's confidence or self-rating of abilities (Marottoli and Richardson, 1998), indicating a lack of awareness of their abilities and limitations. Another possible explanation would be that this group of drivers reflects greater comparative optimism, as reflected in the study by Gosselin et al.

Table 4
Data of optical parameters and the disturbance index for older drivers without and with cataracts. Standard deviation included.

	Strehl ratio mean ± SD (range)	OSI mean ± SD (range)	MTF cut-off (cpd) mean ± SD (range)	Disturbance index mean ± SD (range)
Older drivers without cataract	0.17 ± 0.05 (0.10–0.28)	1.15 ± 0.51 (0.30–2.30)	29.67 ± 10.73 (14.35–51.68)	0.23 ± 0.08 (0.13–0.42)
Older drivers with cataract	0.12 ± 0.05 (0.04–0.25)	2.91 ± 1.55 (0.5–7.0)	18.56 ± 10.86 (7.23–53.02)	0.49 ± 0.25 (0.14–0.87)
Mean comparison (p-value)	$p < 0.001$	$p < 0.001$	$p < 0.001$	0.010

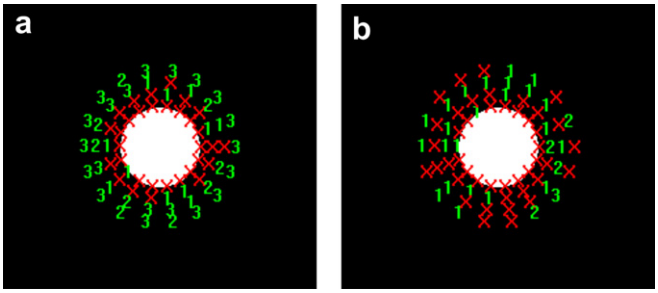


Fig. 3. Graphs made with the Halo v1.0 software for: (a) an older driver without cataracts ($\rho_q = 0.42$), (b) an older driver with cataracts ($\rho_q = 0.81$).

(2010) for three age groups of drivers. These researchers define comparative optimism (CO) as the tendency of individuals to estimate that they are less susceptible to risks than others. Their results indicated that drivers aged 65 and above exhibited a statistically significant level of CO when they considered the risks related to driving, instead of reflecting a cautious attitude. However, a number of studies have reported that older subjects with and without visual impairment had significantly poorer driving performance than younger subjects (Dobbs et al., 1998; Szlyk et al., 1995; Wood, 1999; Wood and Mallon, 2001). Therefore, it has been suggested that drivers 70 and over have more traffic accidents with more severe outcomes per mile driven than do younger drivers (Ryan et al., 1998).

For older drivers affected with cataracts, our results show a higher quantity of aberrations and intraocular scattering, thus affecting retinal-image quality. It is well known that aberrations and ocular scattering result in a lower discrimination capacity and a lower binocular CSF, deteriorating the visual performance (Artal et al., 1993; Fujikado et al., 2004; Jiménez et al., 2008a,b) (Tables 4 and 5 and Figs. 3 and 4). This deterioration in visual performance due to cataract formation could lower driver safety and exacerbate driving difficulties, elevating the accident risk. In the interview, drivers with cataracts showed more difficulty in four situations commonly encountered in nonrural areas: parallel parking, driving in heavy traffic, rush hour driving, and night-time driving (Table 6). Despite that the results of this study clearly reflect that the drivers with cataracts had poorer visual performance, in the questionnaire, we found no differences between cataract and

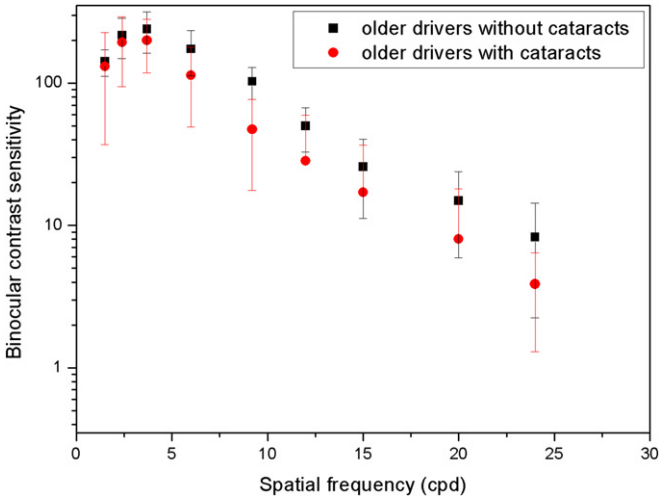


Fig. 4. Average binocular CSF for 15 healthy older drivers and 15 cataract drivers. Data include standard error.

Table 5
Data for the average of 15 older drivers without cataracts and 15 older drivers with the pathology corresponding to binocular CSF.

Spatial frequencies (cpd)	Older drivers without cataract (CSF mean ± SD)	Older drivers with cataract (CSF mean ± SD)	*p*-Value
1.5	131.31 ± 30.31	121.74 ± 94.43	0.486
2.4	217.39 ± 68.09	193.48 ± 98.96	0.447
3.7	239.57 ± 77.94	199.57 ± 81.73	0.181
6.0	173.48 ± 61.05	113.83 ± 64.50	0.015
9.2	102.83 ± 26.16	47.29 ± 29.70	0.000
12.0	49.89 ± 17.15	28.43 ± 31.05	0.026
15.0	25.82 ± 14.59	17.06 ± 19.70	0.177
20.0	14.91 ± 8.99	8.05 ± 10.05	0.059
24.0	8.28 ± 6.03	3.87 ± 2.57	0.027

cataract-free drivers with respect to difficulty in driving in the rain, alone, on freeways, or in making left turns across oncoming traffic. Most older drivers with visual impairment avoid these driving situations, reflecting a safety strategy (Ball et al., 1998; Owsley et al., 1999, 2001). However, it is important to point out that in our study, the subjects with cataracts had good binocular visual acuity (equal to or better than 20/25), and therefore might not avoid challenging driving situations because they might not be aware of their own limitations, thus making driving less safe. Our results support those of other studies indicating that contrast sensitivity and other visual functions such as visual acuity in glare are considerably decreased in older drivers with cataracts (Babizhayev, 2003; Mäntyjärvi and Tuppurainen, 1999), creating a risk factor in traffic and increasing crash risk (Babizhayev, 2003; Owsley et al., 1999). According to the results of the present study, the drivers affected with cataracts should be carefully evaluated, even if visual acuity is sufficient to obtain or renew driver's license because they suffer a deterioration in other important visual functions that can impede the safe performance of activities such as driving.

Others authors have found that older drivers with cataracts are more likely to stop driving compared with those without cataract (Owsley et al., 2001) or to reduce their driving exposure (Owsley et al., 1999). However, when the binocular visual acuity is sufficient to be issued a driver's license, drivers with cataracts may be often unaware of their visual impairment and how it may affect driving performance, until the pathology reaches an advanced stage, which could compromise driving safety. For this reason, it is advisable to explain the visual test results to the subjects, showing them for example Fig. 3 and making them aware of their visual limitations and the implications involved. In view of our results, the measurement of visual acuity may not be a sufficient measure for an applicant to earn or renew a driver's license, since visual performance would not be completely characterized, making it important to examine other visual functions such as contrast sensitivity or halometry. In visual performance, there are other functions that can affect driving safety, such as contrast sensitivity, dark adaptation, colour vision, glare sensitivity and visual field (Mäntyjärvi and Tuppurainen, 1999; Mäntyjärvi et al., 1999; Rogé and Pébayle, 2009; Van Rijn et al., 2011).

Table 6
Driving-difficulty questionnaire for older subjects with and without cataracts. Data presented as percentages.

Driving habits questionnaire	Cataract (% of group)	No cataract (% of group)	*p*-value
1. Driving in the rain			
Difficulty	47	20	0.130
No difficulty	53	80	
2. Driving alone			
Difficulty	20	0	0.072
No difficulty	80	100	
3. Parallel parking			
Difficulty	33	0	0.013
No difficulty	67	100	
4. Left turns in traffic			
Difficulty	33	13	0.208
No difficulty	67	87	
5. Driving on freeways			
Difficulty	13	0	0.153
No difficulty	87	100	
6. Driving in heavy traffic			
Difficulty	60	20	0.025
No difficulty	40	80	
7. Driving in rush hour			
Difficulty	64	20	0.014
No difficulty	36	80	
8. Driving at night			
Difficulty	93	53	0.012
No difficulty	7	47	

In the present study, we have evaluated an important aspect of vision, the visual-discrimination capacity under low-illumination conditions. Our results indicate that elderly people, especially those affected by any ocular disease such as cataract, have a reduced ability to detect peripheral stimuli around the high-luminance stimulus, limiting visual performance. Driving is a highly visual task and therefore it is fundamental to make a complete evaluation of visual performance, especially older drivers. This capacity is key, for example, when driving in the dark, where failure to detect peripheral stimuli around headlights of the oncoming traffic (e.g., pedestrians crossing the street, or any sign on the road) could lead to a traffic accident. Some studies have shown that older drivers have more severe traffic accidents per mile driven than younger drivers (Ryan et al., 1998) and older drivers with cataracts have an elevated crash risk compared to those without pathology (Owsley et al., 1999). Therefore, the visual deterioration that occurs with the normal aging process, as well as eye diseases such as cataract, can notably diminish the ability to perform tasks such as driving, lowering the safety of drivers. Our recommendation is to examine other visual functions such as visual-discrimination capacity, even when binocular visual acuity is sufficient to earn a driver's, at least for older drivers.

Recently, Castro et al. (2011) showed that the Halo software is a useful tool for the study, monitoring, and early diagnosis of different pathologies, such as age-related macular degeneration and keratitis, pathologies that deteriorate the retinal-image quality and visual performance, reflecting a reduction in functions such as the contrast-sensitivity function (Jiménez et al., 2008a,b, 2009). In addition, this software could be useful in patients of laser refractive surgery who frequently mention night-vision disturbances such as a halo around a central source, indicating that they have greater difficulty in discriminating peripheral lights surrounding the central source (Anera et al., 2011; Gutiérrez et al., 2003; Jiménez et al., 2006).

Others works (Babizhayev et al., 2009) have examined the visual impairment using a halometer test in older drivers, but without evaluating the optical quality with an objective method, this being important to determine the deterioration in visual performance. In addition, the study by Babizhayev et al. (2009) showed that the evaluation of night-vision disturbances required hardware, and the use of which could be a disadvantage in clinical examination vs. the Halo v1.0 software used in the present work, which is a free-ware program, enabling its use by any examiner who has a PC, without any additional hardware. On the other hand, it also permits the all parameters to be modified, depending on the type of patient or the experimental conditions, making it possible to customize the test for a particular case or a group of patients. Furthermore, the results found in previous studies have shown the effectiveness of

the Halo software by a significant correlation between the objective measurements and the visual disturbances perceived by the subject (Anera et al., 2011; Castro et al., 2011).

One of the limitations of the present study is that the central stimulus used in the discrimination capacity is a stationary constant stimulation and is relatively weak compared with the headlight of an oncoming vehicle. However, Gutiérrez et al. (2003) using a halometer found that average values for the disturbance index did not depend on the intensity of the central source. It is important to emphasize that all older drivers with and without visual impairment included in this study had difficulty detecting peripheral stimuli around the central one shown with the Halo test, revealing statistically significant differences with respect to other groups of drivers. Therefore, it was expected that a stimulus with greater luminance would provoke more night-vision disturbances (glare or halos), interfering even more with the discrimination of peripheral stimuli, such as pedestrians, cyclists and traffic signs, particularly in older drivers. On the other hand, although the present study lacks an objective driving evaluation (on-road driving test or simulated driving test), we have used a subjective Driving Habits Questionnaire (DHQ), very frequently used in these types of works (Babizhayev et al., 2009; Ball et al., 1998; McGwin et al., 2000), to compare our results on visual function of drivers with their own subjective opinion.

In summary, our results indicate that, older drivers with or without ocular pathology have lower contrast sensitivity and a less ability to detect peripheral stimuli (reduced discrimination capacity) compared with other groups of drivers, even when the binocular visual acuity is sufficient to obtain or renew a driver's license. The discrimination capacity is a crucial visual function for the safe performance of such tasks as driving, where it is fundamental to detect peripheral stimuli, such as pedestrians, bicyclists, or traffic signals. Therefore, although for obtaining or renewing a driver's license, the binocular visual acuity required is 20/40 or better, even if exist an eye disease such as cataract, measurement of visual acuity alone might not be sufficient measures for the issuance or renewal of a license because other visual functions can be considerably decreased, hampering the driving ability, particularly at night (Babizhayev, 2003; Mäntyjärvi and Tuppurainen, 1999). A detailed eye examination is necessary for the evaluation of the ocular health for driving. More studies are needed to set the visual standards for different visual aspects and environments in driving (http://www.icoph.org/downloads/visionfordriving.pdf), although currently the Halo programme acts as a useful tool, being reliable, quick, and easy to perform, and available to any examiner. This tool evaluates the influence of night-vision disturbances such as halos and quantifies the discrimination capacity. Furthermore, it shows the subjects their own visual results, making them aware of their visual limitations and the implications involved. Finally, given the quantitative index provided, it is possible to study the temporal evolution of the disease and also enables a clinician to characterize the degree to which visual performance is affected by the different types of cataracts. Therefore, this test would be advisable in clinical practice for driver's license requirements, particularly for older drivers, evaluating visual functions that, when deteriorated, compromise driving safety.

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